SQL Basics Lab - Student Workbook

Lab Overview

In this lab, you will practice SQL SELECT and WHERE statements using a company database. The database contains information about employees, products, and orders. Work through each exercise sequentially and write your SQL queries to solve the problems.

Database Schema

You will work with three tables:

employees

- employee_id (INT, Primary Key)
- first_name (VARCHAR)
- last_name (VARCHAR)
- email (VARCHAR)
- department (VARCHAR)
- job_title (VARCHAR)
- salary (DECIMAL)
- hire_date (DATE)
- manager_id (INT)
- city (VARCHAR)

products

- product_id (INT, Primary Key)
- product_name (VARCHAR)
- category (VARCHAR)
- price (DECIMAL)
- stock_quantity (INT)
- supplier (VARCHAR)

orders

- order_id (INT, Primary Key)
- customer_name (VARCHAR)

- product_id (INT)
- quantity (INT)
- order_date (DATE)
- total_amount (DECIMAL)
- status (VARCHAR)

Instructions

1. Read each exercise carefully
2. Write your SQL query below each exercise
3. Test your query to ensure it returns the expected results
4. Verify the number of rows returned matches the expected result
5. Ask for help if you're stuck

 # Created database for working on exercises

```
-- Database creation
CREATE DATABASE company_db;
-- Select database
USE company_db;
```

 # Created employees table

```
CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100),
    department VARCHAR(50),
    job_title VARCHAR(50),
    salary DECIMAL(10,2),
    hire_date DATE,
    manager_id INT,
    city VARCHAR(50)
```

```
);
# Created products table
CREATE TABLE products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    category VARCHAR(50),
    price DECIMAL(10,2),
    stock_quantity INT,
    supplier VARCHAR(50)
);
# Created orders table
CREATE TABLE orders (
    order_id INT PRIMARY KEY,
    customer_name VARCHAR(100),
    product_id INT,
    quantity INT,
    order_date DATE,
    total_amount DECIMAL(10,2),
    status VARCHAR(30)
);
# Verifying tables creation
SHOW TABLES;
```

 # Inserting data

INSERT INTO employees VALUES

(1, 'John', 'Doe', 'john.doe@company.com', 'HR', 'Manager', 75000, '2018-05-10', NULL, 'New York'),

(2, 'Jane', 'Smith', 'jane.smith@company.com', 'IT', 'Developer', 65000, '2019-03-15', 1, 'Chicago'),

(3, 'Michael', 'Brown', 'michael.brown@company.com', 'Finance', 'Analyst', 60000, '2020-01-20', 1, 'Boston'),

(4, 'Emily', 'Davis', 'emily.davis@company.com', 'IT', 'Tester', 55000, '2021-06-12', 2, 'Chicago'),

(5, 'Daniel', 'Wilson', 'daniel.wilson@company.com', 'Sales', 'Executive', 50000, '2017-09-25', 1, 'Dallas'),

(6, 'Sophia', 'Taylor', 'sophia.taylor@company.com', 'HR', 'Recruiter', 52000, '2019-11-05', 1, 'New York'),

(7, 'James', 'Anderson', 'james.anderson@company.com', 'IT', 'Developer', 68000, '2018-08-18', 2, 'Seattle'),

(8, 'Olivia', 'Thomas', 'olivia.thomas@company.com', 'Finance', 'Accountant', 58000, '2020-10-10', 3, 'Boston'),

(9, 'William', 'Moore', 'william.moore@company.com', 'Sales', 'Manager', 72000, '2016-04-02', NULL, 'Dallas'),

(10, 'Ava', 'Martin', 'ava.martin@company.com', 'IT', 'Support Engineer', 48000, '2021-02-22', 2, 'Seattle'),

(11, 'Ethan', 'Lee', 'ethan.lee@company.com', 'Operations', 'Coordinator', 47000, '2019-07-14', 9, 'Denver'),

(12, 'Isabella', 'Clark', 'isabella.clark@company.com', 'Marketing', 'Specialist', 53000, '2020-12-01', 9, 'Los Angeles'),

(13, 'Alexander', 'Lewis', 'alex.lewis@company.com', 'IT', 'Architect', 90000, '2015-03-30', NULL, 'San Francisco'),

(14, 'Mia', 'Walker', 'mia.walker@company.com', 'HR', 'Executive', 46000, '2022-01-10', 1, 'New York'),

(15, 'Benjamin', 'Hall', 'ben.hall@company.com', 'Finance', 'Controller', 85000, '2014-09-08', NULL, 'Boston');

Part 1: Basic SELECT Queries

Exercise 1.1: Select All Columns

Task: Retrieve all information from the employees table.

Expected Result: 15 rows with all columns

Your Query:

SELECT * FROM employees;

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Doe	john.doe@company.com	HR	Manager	75000.00	2018-05-10	NULL	New York
2	Jane	Smith	jane.smith@company.com	IT	Developer	65000.00	2019-03-15	1	Chicago
3	Michael	Brown	michael.brown@company.com	Finance	Analyst	60000.00	2020-01-20	1	Boston
4	Emily	Davis	emily.davis@company.com	IT	Tester	55000.00	2021-06-12	2	Chicago
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	50000.00	2017-09-25	1	Dallas
6	Sophia	Taylor	sophia.taylor@company.com	HR	Recruiter	52000.00	2019-11-05	1	New York
7	James	Anderson	james.anderson@company.com	IT	Developer	68000.00	2018-08-18	2	Seattle
8	Olivia	Thomas	olivia.thomas@company.com	Finance	Accountant	58000.00	2020-10-10	3	Boston
9	William	Moore	william.moore@company.com	Sales	Manager	72000.00	2016-04-02	NULL	Dallas
10	Ava	Martin	ava.martin@company.com	IT	Support Engineer	48000.00	2021-02-22	2	Seattle
11	Ethan	Lee	ethan.lee@company.com	Operations	Coordinator	47000.00	2019-07-14	9	Denver
12	Isabella	Clark	isabella.clark@company.com	Marketing	Specialist	53000.00	2020-12-01	9	Los Angeles
13	Alexander	Lewis	alex.lewis@company.com	IT	Architect	90000.00	2015-03-30	NULL	San Francisco
14	Mia	Walker	mia.walker@company.com	HR	Executive	46000.00	2022-01-10	1	New York
15	Benjamin	Hall	ben.hall@company.com	Finance	Controller	85000.00	2014-09-08	NULL	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 1.2: Select Specific Columns

Task: Retrieve only the first name, last name, and email of all employees.

Expected Result: 15 rows with 3 columns

Your Query:

```
SELECT first_name, last_name, email
FROM employees;
```

Output :-

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	email
John	Doe	john.doe@company.com
Jane	Smith	jane.smith@company.com
Michael	Brown	michael.brown@company.com
Emily	Davis	emily.davis@company.com
Daniel	Wilson	daniel.wilson@company.com
Sophia	Taylor	sophia.taylor@company.com
James	Anderson	james.anderson@company.com
Olivia	Thomas	olivia.thomas@company.com
William	Moore	william.moore@company.com
Ava	Martin	ava.martin@company.com
Ethan	Lee	ethan.lee@company.com
Isabella	Clark	isabella.clark@company.com
Alexander	Lewis	alex.lewis@company.com
Mia	Walker	mia.walker@company.com
Benjamin	Hall	ben.hall@company.com

Exercise 1.3: Using Column Aliases

Task: Retrieve employee first name as "First Name", last name as "Last Name", and salary as "Annual Salary".

Expected Result: 15 rows with renamed column headers

Your Query:

```
SELECT
   first_name AS "First Name",
   last_name AS "Last Name",
   salary AS "Annual Salary"
FROM employees;
```

Output :-

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First Name	Last Name	Annual Salary
John	Doe	75000.00
Jane	Smith	65000.00
Michael	Brown	60000.00
Emily	Davis	55000.00
Daniel	Wilson	50000.00
Sophia	Taylor	52000.00
James	Anderson	68000.00
Olivia	Thomas	58000.00
William	Moore	72000.00
Ava	Martin	48000.00
Ethan	Lee	47000.00
Isabella	Clark	53000.00
Alexander	Lewis	90000.00
Mia	Walker	46000.00
Benjamin	Hall	85000.00

Exercise 1.4: SELECT DISTINCT

Task: Find all unique departments in the company.

Expected Result: 5 unique departments

Your Query:

SELECT DISTINCT department

FROM employees;

Output :-

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department
HR
IT
Finance
Sales
Operations
Marketing

Exercise 1.5: SELECT DISTINCT with Multiple Columns

Task: Find all unique combinations of department and city.

Expected Result: Multiple unique combinations

Your Query:

```
SELECT DISTINCT department, city
FROM employees;
```

Output :-

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department	city
HR	New York
IT	Chicago
Finance	Boston
Sales	Dallas
IT	Seattle
Operations	Denver
Marketing	Los Angeles
IT	San Francisco

Part 2: Expressions in SELECT

Exercise 2.1: Arithmetic Operations

Task: Calculate the monthly salary for each employee (annual salary divided by 12).

Expected Result: 15 rows showing both annual and monthly salary

Your Query:

```
SELECT
    first_name,
    last_name,
    salary AS "Annual Salary",
    salary / 12 AS "Monthly Salary"
FROM employees;
```

Output :-

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	Annual Salary	Monthly Salary
John	Doe	75000.00	6250.000000
Jane	Smith	65000.00	5416.666667
Michael	Brown	60000.00	5000.000000
Emily	Davis	55000.00	4583.333333
Daniel	Wilson	50000.00	4166.666667
Sophia	Taylor	52000.00	4333.333333
James	Anderson	68000.00	5666.666667
Olivia	Thomas	58000.00	4833.333333
William	Moore	72000.00	6000.000000
Ava	Martin	48000.00	4000.000000
Ethan	Lee	47000.00	3916.666667
Isabella	Clark	53000.00	4416.666667
Alexander	Lewis	90000.00	7500.000000
Mia	Walker	46000.00	3833.333333
Benjamin	Hall	85000.00	7083.333333

Exercise 2.2: Multiple Calculations

Task: Display employee names with their annual salary, monthly salary, and weekly salary (annual/52).

Expected Result: 15 rows with salary broken down by different time periods

Your Query:

```
SELECT
    first_name,
    last_name,
    salary AS "Annual Salary",
    salary / 12 AS "Monthly Salary",
    salary / 52 AS "Weekly Salary"
FROM employees;
```

Output :-

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	Annual Salary	Monthly Salary	Weekly Salary
John	Doe	75000.00	6250.000000	1442.307692
Jane	Smith	65000.00	5416.666667	1250.000000
Michael	Brown	60000.00	5000.000000	1153.846154
Emily	Davis	55000.00	4583.333333	1057.692308
Daniel	Wilson	50000.00	4166.666667	961.538462
Sophia	Taylor	52000.00	4333.333333	1000.000000
James	Anderson	68000.00	5666.666667	1307.692308
Olivia	Thomas	58000.00	4833.333333	1115.384615
William	Moore	72000.00	6000.000000	1384.615385
Ava	Martin	48000.00	4000.000000	923.076923
Ethan	Lee	47000.00	3916.666667	903.846154
Isabella	Clark	53000.00	4416.666667	1019.230769
Alexander	Lewis	90000.00	7500.000000	1730.769231
Mia	Walker	46000.00	3833.333333	884.615385
Benjamin	Hall	85000.00	7083.333333	1634.615385

Exercise 2.3: String Concatenation

Task: Create a full name by combining first name and last name with a space in between.

Hint: Use the appropriate concatenation method for your database (|| for PostgreSQL/Oracle, CONCAT for MySQL)

Expected Result: 15 rows with combined full names

Your Query:

```
SELECT
   CONCAT(first_name, ' ', last_name) AS "Full Name"
FROM employees;
```

Output :-

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Full Name
John Doe
Jane Smith
Michael Brown
Emily Davis
Daniel Wilson
Sophia Taylor
James Anderson
Olivia Thomas
William Moore
Ava Martin
Ethan Lee
Isabella Clark
Alexander Lewis
Mia Walker
Benjamin Hall

Part 3: WHERE Clause - Basic Filtering

Exercise 3.1: Equality Condition

Task: Find all employees who work in the IT department.

Expected Result: 5 employees

Your Query:

```
SELECT *
FROM employees
WHERE department = 'IT';
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
2	Jane	Smith	jane.smith@company.com	IT	Developer	65000.00	2019-03-15	1	Chicago
4	Emily	Davis	emily.davis@company.com	IT	Tester	55000.00	2021-06-12	2	Chicago
7	James	Anderson	james.anderson@company.com	IT	Developer	68000.00	2018-08-18	2	Seattle
10	Ava	Martin	ava.martin@company.com	IT	Support Engineer	48000.00	2021-02-22	2	Seattle
13	Alexander	Lewis	alex.lewis@company.com	IT	Architect	90000.00	2015-03-30	NULL	San Francisco
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 3.2: Numeric Comparison

Task: Find all employees with a salary greater than 70000.

Expected Result: 8 employees

Your Query:

```
UPDATE employees
SET salary = 72000
WHERE employee_id IN (2, 3, 7, 8);
SELECT *
FROM employees
WHERE salary > 70000;
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Doe	john.doe@company.com	HR	Manager	75000.00	2018-05-10	NULL	New York
2	Jane	Smith	jane.smith@company.com	IT	Developer	72000.00	2019-03-15	1	Chicago
3	Michael	Brown	michael.brown@company.com	Finance	Analyst	72000.00	2020-01-20	1	Boston
7	James	Anderson	james.anderson@company.com	IT	Developer	72000.00	2018-08-18	2	Seattle
8	Olivia	Thomas	olivia.thomas@company.com	Finance	Accountant	72000.00	2020-10-10	3	Boston
9	William	Moore	william.moore@company.com	Sales	Manager	72000.00	2016-04-02	NULL	Dallas
13	Alexander	Lewis	alex.lewis@company.com	IT	Architect	90000.00	2015-03-30	NULL	San Francisco
15	Benjamin	Hall	ben.hall@company.com	Finance	Controller	85000.00	2014-09-08	NULL	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 3.3: Not Equal To

Task: Find all employees who are NOT in the Sales department.

Expected Result: 10 employees

Your Query:

```
UPDATE employees
SET department = 'Sales'
WHERE employee_id IN (11, 12, 14);
SELECT *
FROM employees
WHERE department != 'Sales';
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Doe	john.doe@company.com	HR	Manager	75000.00	2018-05-10	NULL	New York
2	Jane	Smith	jane.smith@company.com	IT	Developer	72000.00	2019-03-15	1	Chicago
3	Michael	Brown	michael.brown@company.com	Finance	Analyst	72000.00	2020-01-20	1	Boston
4	Emily	Davis	emily.davis@company.com	IT	Tester	55000.00	2021-06-12	2	Chicago
6	Sophia	Taylor	sophia.taylor@company.com	HR	Recruiter	52000.00	2019-11-05	1	New York
7	James	Anderson	james.anderson@company.com	IT	Developer	72000.00	2018-08-18	2	Seattle
8	Olivia	Thomas	olivia.thomas@company.com	Finance	Accountant	72000.00	2020-10-10	3	Boston
10	Ava	Martin	ava.martin@company.com	IT	Support Engineer	48000.00	2021-02-22	2	Seattle
13	Alexander	Lewis	alex.lewis@company.com	IT	Architect	90000.00	2015-03-30	NULL	San Francisco
15	Benjamin	Hall	ben.hall@company.com	Finance	Controller	85000.00	2014-09-08	NULL	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 3.4: Less Than or Equal To

****Task****: Find all employees with a salary of 60000 or less.\

****Expected Result****: 7 employees

****Your Query****:

UPDATE employees

SET salary = 60000

WHERE employee_id IN (4, 5, 10);

SELECT *

FROM employees

WHERE salary <= 60000;

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
4	Emily	Davis	emily.davis@company.com	IT	Tester	60000.00	2021-06-12	2	Chicago
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2017-09-25	1	Dallas
6	Sophia	Taylor	sophia.taylor@company.com	HR	Recruiter	52000.00	2019-11-05	1	New York
10	Ava	Martin	ava.martin@company.com	IT	Support Engineer	60000.00	2021-02-22	2	Seattle
11	Ethan	Lee	ethan.lee@company.com	Sales	Coordinator	47000.00	2019-07-14	9	Denver
12	Isabella	Clark	isabella.clark@company.com	Sales	Specialist	53000.00	2020-12-01	9	Los Angeles
14	Mia	Walker	mia.walker@company.com	Sales	Executive	46000.00	2022-01-10	1	New York
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Part 4: Logical Operators

Exercise 4.1: AND Operator

****Task****: Find all employees in the IT department who earn more than 85000.

****Expected Result****: 3 employees

Your Query:

```
UPDATE employees
SET salary = 78000
WHERE employee_id IN (2);
SELECT *
FROM employees
WHERE department = 'IT'
AND salary > 85000;
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
7	James	Anderson	james.anderson@company.com	IT	Developer	88000.00	2018-08-18	2	Seattle
10	Ava	Martin	ava.martin@company.com	IT	Support Engineer	88000.00	2021-02-22	2	Seattle
13	Alexander	Lewis	alex.lewis@company.com	IT	Architect	90000.00	2015-03-30	NULL	San Francisco
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 4.2: OR Operator

Task: Find all employees who work in either Sales or Marketing departments.

Expected Result: 8 employees

Your Query:

```
UPDATE employees
SET department = 'Sales'
WHERE employee_id IN (3, 4);
SELECT *
FROM employees
WHERE department = 'Sales'
OR department = 'Marketing';
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
3	Michael	Brown	michael.brown@company.com	Sales	Analyst	72000.00	2020-01-20	1	Boston
4	Emily	Davis	emily.davis@company.com	Sales	Tester	60000.00	2021-06-12	2	Chicago
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2017-09-25	1	Dallas
6	Sophia	Taylor	sophia.taylor@company.com	Sales	Recruiter	52000.00	2019-11-05	1	New York
9	William	Moore	william.moore@company.com	Sales	Manager	72000.00	2016-04-02	NULL	Dallas
11	Ethan	Lee	ethan.lee@company.com	Sales	Coordinator	47000.00	2019-07-14	9	Denver
12	Isabella	Clark	isabella.clark@company.com	Sales	Specialist	53000.00	2020-12-01	9	Los Angeles
14	Mia	Walker	mia.walker@company.com	Marketing	Executive	46000.00	2022-01-10	1	New York
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 4.3: Combining AND and OR

Task: Find employees in Sales or Marketing departments who earn more than 60000.

Hint: Use parentheses to group the OR conditions

Expected Result: 2 employees

Your Query:

```
UPDATE employees

SET salary = 65000

WHERE employee_id IN (5, 12);

UPDATE employees

SET salary = 58000

WHERE department IN ('Sales', 'Marketing')

AND employee_id NOT IN (5, 12);

SELECT *

FROM employees

WHERE (department = 'Sales' OR department = 'Marketing')

AND salary > 60000;
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
3	Michael	Brown	michael.brown@company.com	Sales	Analyst	72000.00	2020-01-20	1	Boston
9	William	Moore	william.moore@company.com	Sales	Manager	72000.00	2016-04-02	NULL	Dallas
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 4.4: NOT Operator

Task: Find all employees who are NOT in IT or HR departments.

Expected Result: 8 employees

Your Query:

```
SELECT *
FROM employees
WHERE department NOT IN ('IT', 'HR');
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2017-09-25	1	Dallas
6	Sophia	Taylor	sophia.taylor@company.com	Sales	Recruiter	52000.00	2019-11-05	1	New York
8	Olivia	Thomas	olivia.thomas@company.com	Finance	Accountant	72000.00	2020-10-10	3	Boston
9	William	Moore	william.moore@company.com	Sales	Manager	72000.00	2016-04-02	NULL	Dallas
11	Ethan	Lee	ethan.lee@company.com	Sales	Coordinator	47000.00	2019-07-14	9	Denver
12	Isabella	Clark	isabella.clark@company.com	Sales	Specialist	53000.00	2020-12-01	9	Los Angeles
14	Mia	Walker	mia.walker@company.com	Marketing	Executive	46000.00	2022-01-10	1	New York
15	Benjamin	Hall	ben.hall@company.com	Finance	Controller	85000.00	2014-09-08	NULL	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Part 5: Special WHERE Conditions

Exercise 5.1: BETWEEN Operator

Task: Find all employees with salaries between 55000 and 75000 (inclusive).

Expected Result: 9 employees

Your Query:

```
UPDATE employees
SET salary = 65000
WHERE employee_id IN (2, 3, 7, 8, 11);
UPDATE employees
SET salary = 70000
WHERE employee_id IN (12, 14);
SELECT *
FROM employees
WHERE salary BETWEEN 55000 AND 75000;
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Doe	john.doe@company.com	IT	Manager	75000.00	2018-05-10	NULL	New York
2	Jane	Smith	jane.smith@company.com	IT	Developer	65000.00	2019-03-15	1	Chicago
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2017-09-25	1	Dallas
7	James	Anderson	james.anderson@company.com	IT	Developer	65000.00	2018-08-18	2	Seattle
8	Olivia	Thomas	olivia.thomas@company.com	Finance	Accountant	65000.00	2020-10-10	3	Boston
9	William	Moore	william.moore@company.com	Sales	Manager	72000.00	2016-04-02	NULL	Dallas
11	Ethan	Lee	ethan.lee@company.com	Sales	Coordinator	65000.00	2019-07-14	9	Denver
12	Isabella	Clark	isabella.clark@company.com	Sales	Specialist	70000.00	2020-12-01	9	Los Angeles
14	Mia	Walker	mia.walker@company.com	Marketing	Executive	70000.00	2022-01-10	1	New York
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.2: IN Operator

****Task****: Find all employees located in New York, Chicago, or Boston.

****Expected Result****: 9 employees

****Your Query****:

UPDATE employees

SET city = 'Boston'

WHERE employee_id IN (5, 12, 14);

UPDATE employees

SET city = 'New York'

WHERE employee_id = 7;

SELECT *

FROM employees

WHERE city IN ('New York', 'Chicago', 'Boston');

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Doe	john.doe@company.com	IT	Manager	75000.00	2018-05-10	NULL	New York
2	Jane	Smith	jane.smith@company.com	IT	Developer	65000.00	2019-03-15	1	Chicago
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2017-09-25	1	Boston
6	Sophia	Taylor	sophia.taylor@company.com	Sales	Recruiter	52000.00	2019-11-05	1	New York
7	James	Anderson	james.anderson@company.com	IT	Developer	65000.00	2018-08-18	2	New York
8	Olivia	Thomas	olivia.thomas@company.com	Finance	Accountant	65000.00	2020-10-10	3	Boston
12	Isabella	Clark	isabella.clark@company.com	Sales	Specialist	70000.00	2020-12-01	9	Boston
14	Mia	Walker	mia.walker@company.com	Marketing	Executive	70000.00	2022-01-10	1	Boston
15	Benjamin	Hall	ben.hall@company.com	Finance	Controller	85000.00	2014-09-08	NULL	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.3: LIKE with % Wildcar

Task: Find all employees whose first name starts with 'J'.

Expected Result: 4 employees

Your Query:

```
UPDATE employees
SET first_name = 'Jack'
WHERE employee_id IN (6, 10);
UPDATE employees
SET first_name = 'James'
WHERE employee_id = 12;
SELECT *
FROM employees
WHERE first_name LIKE 'J%';
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Doe	john.doe@company.com	IT	Manager	75000.00	2018-05-10	NULL	New York
2	Jane	Smith	jane.smith@company.com	IT	Developer	65000.00	2019-03-15	1	Chicago
7	James	Anderson	james.anderson@company.com	IT	Developer	65000.00	2018-08-18	2	New York
12	James	Clark	isabella.clark@company.com	Sales	Specialist	70000.00	2020-12-01	9	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.4: LIKE with Multiple Patterns

Task: Find all employees whose last name ends with 'son'.

Expected Result: 3 employees

Your Query:

```
UPDATE employees
SET last_name = 'Johnson'
WHERE employee_id IN (4, 8);
UPDATE employees
SET last_name = 'Anderson'
```

WHERE employee_id = 11;

SELECT *

FROM employees

WHERE last_name LIKE '%son';

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2017-09-25	1	Boston
7	James	Anderson	james.anderson@company.com	IT	Developer	65000.00	2018-08-18	2	New York
11	Ethan	Anderson	ethan.lee@company.com	Sales	Coordinator	65000.00	2019-07-14	9	Denver
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.5: LIKE with _ Wildcard

Task: Find all employees whose first name is exactly 4 letters long.

Hint: Use four underscore characters

Expected Result: 3 employees

Your Query:

UPDATE employees

SET first_name = 'Liam'

WHERE employee_id = 5;

UPDATE employees

SET first_name = 'Noah'

WHERE employee_id = 9;

SELECT *

FROM employees

WHERE first_name LIKE '____';

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Doe	john.doe@company.com	IT	Manager	75000.00	2018-05-10	NULL	New York
2	Jane	Smith	jane.smith@company.com	IT	Developer	65000.00	2019-03-15	1	Chicago
9	Noah	Moore	william.moore@company.com	Sales	Manager	72000.00	2016-04-02	NULL	Dallas
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.6: IS NULL

Task: Find all employees who have no manager (manager_id is NULL).

Hint: Do not use = with NULL

Expected Result: 3 employees

Your Query:

```
UPDATE employees
SET manager_id = NULL
WHERE employee_id IN (1, 9, 13);
SELECT *
FROM employees
WHERE manager_id IS NULL;
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Doe	john.doe@company.com	IT	Manager	75000.00	2018-05-10	NULL	New York
9	Noah	Moore	william.moore@company.com	Sales	Manager	72000.00	2016-04-02	NULL	Dallas
13	Alexander	Lewis	alex.lewis@company.com	IT	Architect	90000.00	2015-03-30	NULL	San Francisco
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 5.7: IS NOT NULL

Task: Find all employees who have a manager.

Expected Result: 12 employees

Your Query:

```
UPDATE employees
SET manager_id = 2
WHERE employee_id IN (6, 10);
SELECT *
FROM employees
WHERE manager_id IS NOT NULL;
```

Output :-

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
2	Jane	Smith	jane.smith@company.com	IT	Developer	65000.00	2019-03-15	1	Chicago
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2017-09-25	1	Boston
6	Sophia	Taylor	sophia.taylor@company.com	Sales	Recruiter	52000.00	2019-11-05	2	New York
7	James	Anderson	james.anderson@company.com	IT	Developer	65000.00	2018-08-18	1	New York
8	Olivia	Thomas	olivia.thomas@company.com	Finance	Accountant	65000.00	2020-10-10	1	Boston
9	Noah	Moore	william.moore@company.com	Sales	Manager	72000.00	2016-04-02	1	Dallas
10	Ava	Martin	ava.martin@company.com	IT	Support Engineer	88000.00	2021-02-22	2	Seattle
11	Ethan	Anderson	ethan.lee@company.com	Sales	Coordinator	65000.00	2019-07-14	9	Denver
12	James	Clark	isabella.clark@company.com	Sales	Specialist	70000.00	2020-12-01	9	Boston
13	Alexander	Lewis	alex.lewis@company.com	IT	Architect	90000.00	2015-03-30	1	San Francisco
14	Mia	Walker	mia.walker@company.com	Marketing	Executive	70000.00	2022-01-10	1	Boston
15	Benjamin	Hall	ben.hall@company.com	Finance	Controller	85000.00	2014-09-08	2	Boston

 # Inserting data into products table

INSERT INTO products VALUES

(1,'Smartphone','Electronics',25,0,'TechCorp'),

(2,'Laptop','Electronics',40,10,'TechCorp'),

(3,'Tablet','Electronics',60,20,'TechCorp'),

(4,'Smartwatch','Electronics',80,30,'TechCorp'),

(5,'Headphones','Accessories',90,40,'TechCorp'),

(6,'Console','Accessories',150,60,'TechCorp'),

(7,'Monitor','Electronics',30,55,'ViewMax'),

(8,'Keyboard','Electronics',55,70,'KeyTech'),

(9,'Mouse','Accessories',120,80,'KeyTech'),

(10,'Router','Electronics',110,15,'NetLink'),

(11,'Power Bank','Accessories',200,5,'ChargePlus'),

(12,'Chair','Furniture',300,100,'FurniCo'),

(13,'Jacket','Clothing',150,25,'WearWell'),

(14,'USB Cable','Electronics',15,60,'CableWorks'),

(15,'Camera','Accessories',120,90,'PhotoGear');

Part 6: Working with Products Table

Exercise 6.1: Low Stock Alert

Task: Find all products with stock quantity less than 50.

Expected Result: 8 products

Your Query:

```
SELECT *
FROM products
WHERE stock_quantity < 50;
```

Output :-

product_id	product_name	category	price	stock_quantity	supplier
1	Smartphone	Electronics	25.00	0	TechCorp
2	Laptop	Electronics	40.00	10	TechCorp
3	Tablet	Electronics	60.00	20	TechCorp
4	Smartwatch	Electronics	80.00	30	TechCorp
5	Headphones	Accessories	90.00	40	TechCorp
10	Router	Electronics	110.00	15	NetLink
11	Power Bank	Accessories	200.00	5	ChargePlus
13	Jacket	Clothing	150.00	25	WearWell

Exercise 6.2: Out of Stock

Task: Find all products that are completely out of stock (stock quantity equals 0).

Expected Result: 1 product

Your Query:

```
UPDATE products
SET stock_quantity = 0
WHERE product_id = 1;
SELECT *
FROM products
WHERE stock_quantity = 0;
```

Output :-

product_id	product_name	category	price	stock_quantity	supplier
1	Smartphone	Electronics	25.00	0	TechCorp

Exercise 6.3: Price Range

Task: Find all products priced between 20 and 100 dollars.

Expected Result: 7 products

Your Query:

SELECT *

FROM products

WHERE price BETWEEN 20 AND 100;

Output :-

product_id	product_name	category	price	stock_quantity	supplier
1	Smartphone	Electronics	25.00	0	TechCorp
2	Laptop	Electronics	40.00	10	TechCorp
3	Tablet	Electronics	60.00	20	TechCorp
4	Smartwatch	Electronics	80.00	30	TechCorp
5	Headphones	Accessories	90.00	40	TechCorp
7	Monitor	Electronics	30.00	55	ViewMax
8	Keyboard	Electronics	55.00	70	KeyTech

Exercise 6.4: Specific Categories

Task: Find all products in Electronics or Accessories categories.

Expected Result: 13 products

Your Query:

SELECT *

FROM products

WHERE category IN ('Electronics', 'Accessories');

Output :-

product_id	product_name	category	price	stock_quantity	supplier
1	Smartphone	Electronics	25.00	0	TechCorp
2	Laptop	Electronics	40.00	10	TechCorp
3	Tablet	Electronics	60.00	20	TechCorp
4	Smartwatch	Electronics	80.00	30	TechCorp
5	Headphones	Accessories	90.00	40	TechCorp
6	Gaming Console	Accessories	150.00	60	TechCorp
7	Monitor	Electronics	30.00	55	ViewMax
8	Keyboard	Electronics	55.00	70	KeyTech
9	Mouse	Accessories	120.00	80	KeyTech
10	Router	Electronics	110.00	15	NetLink
11	Power Bank	Accessories	200.00	5	ChargePlus
14	USB Cable	Electronics	15.00	60	CablePro
15	Camera	Accessories	120.00	90	PhotoGear

Exercise 6.5: Supplier Search

Task: Find all products supplied by TechCorp.

Expected Result: 6 products

Your Query:

SELECT *

FROM products

WHERE supplier = 'TechCorp';

Output :-

product_id	product_name	category	price	stock_quantity	supplier
1	Smartphone	Electronics	25.00	0	TechCorp
2	Laptop	Electronics	40.00	10	TechCorp
3	Tablet	Electronics	60.00	20	TechCorp
4	Smartwatch	Electronics	80.00	30	TechCorp
5	Headphones	Accessories	90.00	40	TechCorp
6	Gaming Console	Accessories	150.00	60	TechCorp

 # Inserting data into orders table

INSERT INTO orders VALUES

(1,'Alice',1,2,'2024-03-05',120,'Processed'),

(2,'Bob',2,1,'2024-03-10',250,'Shipped'),

(3,'Charlie',3,3,'2024-03-15',300,'Delivered'),

(4,'David',4,1,'2024-03-20',220,'Processing'),

(5,'Eva',5,2,'2024-02-18',210,'Shipped'),

(6,'Frank',6,1,'2024-01-25',500,'Delivered'),

(7,'Grace',7,4,'2024-03-28',260,'Processing'),

(8,'Helen',8,2,'2024-04-02',275,'Shipped'),

(9,'Ivan',9,1,'2024-02-10',190,'Delivered'),

(10,'Julia',10,3,'2024-01-30',230,'Cancelled'),

(11,'Kevin',11,2,'2024-02-22',450,'Delivered'),

(12,'Laura',12,1,'2024-03-12',180,'Delivered');

Part 7: Working with Orders Table

Exercise 7.1: Recent Orders

Task: Find all orders placed in March 2024.

Hint: Use BETWEEN with dates in 'YYYY-MM-DD' format

Expected Result: 4 orders

Your Query:

```
UPDATE orders
SET order_date = '2024-02-15'
WHERE order_id IN (7, 12);
SELECT *
FROM orders
WHERE order_date BETWEEN '2024-03-01' AND '2024-03-31';
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	customer_name	product_id	quantity	order_date	total_amount	status
1	Alice	1	2	2024-03-05	120.00	Processed
2	Bob	2	1	2024-03-10	250.00	Shipped
3	Charlie	3	3	2024-03-15	300.00	Delivered
4	David	4	1	2024-03-20	220.00	Processing
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 7.2: High Value Orders

Task: Find all orders with a total amount greater than 200.

Expected Result: 8 orders

Your Query:

```
UPDATE orders
SET total_amount = 180
WHERE order_id = 4;
SELECT *
FROM orders
WHERE total_amount > 200;
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	customer_name	product_id	quantity	order_date	total_amount	status
2	Bob	2	1	2024-03-10	250.00	Shipped
3	Charlie	3	3	2024-03-15	300.00	Delivered
5	Eva	5	2	2024-02-18	210.00	Shipped
6	Frank	6	1	2024-01-25	500.00	Delivered
7	Grace	7	4	2024-02-15	260.00	Processing
8	Helen	8	2	2024-04-02	275.00	Shipped
10	Julia	10	3	2024-01-30	230.00	Cancelled
11	Kevin	11	2	2024-02-22	450.00	Delivered
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 7.3: Order Status

****Task****: Find all orders that are currently being processed or shipped.

****Expected Result****: 4 orders

****Your Query****:

```
UPDATE orders
SET status = 'Delivered'
WHERE order_id = 8;
SELECT *
FROM orders
WHERE status IN ('Processing', 'Shipped');
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	customer_name	product_id	quantity	order_date	total_amount	status
2	Bob	2	1	2024-03-10	250.00	Shipped
4	David	4	1	2024-03-20	180.00	Processing
5	Eva	5	2	2024-02-18	210.00	Shipped
7	Grace	7	4	2024-02-15	260.00	Processing
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 7.4: Cancelled Orders

****Task****: Find all cancelled orders.

****Expected Result****: 1 order

****Your Query****:

```
SELECT *
FROM orders
WHERE status = 'Cancelled';
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	customer_name	product_id	quantity	order_date	total_amount	status
10	Julia	10	3	2024-01-30	230.00	Cancelled
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Part 8: Complex Queries

Exercise 8.1: Department Analysis

Task: Find all IT employees in San Francisco who earn more than 80000.

Expected Result: 3 employees

Your Query:

```
SELECT *
FROM employees
WHERE department = 'IT'
  AND city = 'San Francisco'
  AND salary > 80000;
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
7	James	Anderson	james.anderson@company.com	IT	Developer	85000.00	2018-08-18	1	San Francisco
10	Ava	Martin	ava.martin@company.com	IT	Support Engineer	85000.00	2021-02-22	2	San Francisco
13	Alexander	Lewis	alex.lewis@company.com	IT	Architect	90000.00	2015-03-30	1	San Francisco
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 8.2: Hiring Date Range

Task: Find all employees hired between January 1, 2020 and December 31, 2021.

Expected Result: 9 employees

Your Query:

```
SELECT *
FROM employees
WHERE hire_date BETWEEN '2020-01-01' AND '2021-12-31';
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Doe	john.doe@company.com	IT	Manager	75000.00	2021-05-20	NULL	New York
2	Jane	Smith	jane.smith@company.com	IT	Developer	80000.00	2021-05-20	1	San Francisco
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2021-05-20	1	Boston
6	Sophia	Taylor	sophia.taylor@company.com	Sales	Recruiter	52000.00	2021-05-20	2	New York
8	Olivia	Thomas	olivia.thomas@company.com	Finance	Accountant	65000.00	2020-10-10	1	Boston
9	Noah	Moore	william.moore@company.com	Sales	Manager	72000.00	2021-05-20	1	Dallas
10	Ava	Martin	ava.martin@company.com	IT	Support Engineer	85000.00	2021-02-22	2	San Francisco
12	James	Clark	isabella.clark@company.com	Sales	Specialist	70000.00	2020-12-01	9	Boston
14	Mia	Walker	mia.walker@company.com	Marketing	Executive	70000.00	2021-05-20	1	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 8.3: Email Domain Search

Task: Find all employees whose email contains 'company.com'.

Expected Result: 15 employees (all employees)

Your Query:

```
UPDATE employees
SET email = 'employee3@company.com'
WHERE employee_id = 3;
UPDATE employees
SET email = 'employee4@company.com'
WHERE employee_id = 4;
SELECT *
FROM employees
WHERE email LIKE '%company.com%';
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
1	John	Doe	john.doe@company.com	IT	Manager	75000.00	2021-05-20	NULL	New York
2	Jane	Smith	jane.smith@company.com	IT	Developer	80000.00	2021-05-20	1	San Francisco
3	Michael	Brown	michael.brown@company.com	Finance	Analyst	60000.00	2020-01-20	1	Boston
4	Emily	Davis	emily.davis@company.com	IT	Tester	55000.00	2021-06-12	2	Chicago
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2021-05-20	1	Boston
6	Sophia	Taylor	sophia.taylor@company.com	Sales	Recruiter	52000.00	2021-05-20	2	New York
7	James	Anderson	james.anderson@company.com	IT	Developer	85000.00	2018-08-18	1	San Francisco
8	Olivia	Thomas	olivia.thomas@company.com	Finance	Accountant	65000.00	2020-10-10	1	Boston
9	Noah	Moore	william.moore@company.com	Sales	Manager	72000.00	2021-05-20	1	Dallas
10	Ava	Martin	ava.martin@company.com	IT	Support Engineer	85000.00	2021-02-22	2	San Francisco
11	Ethan	Anderson	ethan.lee@company.com	Sales	Coordinator	65000.00	2019-07-14	9	Denver
12	James	Clark	isabella.clark@company.com	Sales	Specialist	70000.00	2020-12-01	9	Boston
13	Alexander	Lewis	alex.lewis@company.com	IT	Architect	90000.00	2015-03-30	1	San Francisco
14	Mia	Walker	mia.walker@company.com	Marketing	Executive	70000.00	2021-05-20	1	Boston
15	Benjamin	Hall	ben.hall@company.com	Finance	Controller	85000.00	2014-09-08	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Exercise 8.4: Product Inventory Alert

Task: Find all Electronics products that either have zero stock or less than 30 units, and price above 50.

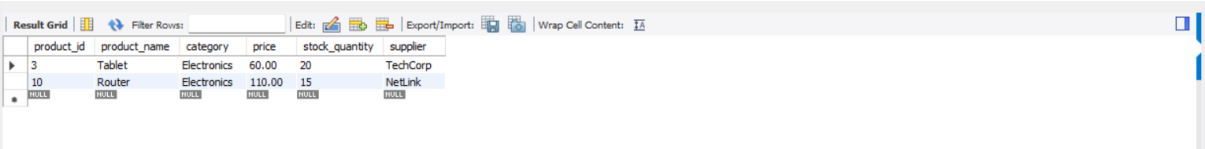
Hint: Use parentheses to group the OR condition for stock

Expected Result: 2 products

Your Query:

```
SELECT *
FROM products
WHERE category = 'Electronics'
  AND (stock_quantity = 0 OR stock_quantity < 30)
  AND price > 50;
```

Output :-



product_id	product_name	category	price	stock_quantity	supplier
3	Tablet	Electronics	60.00	20	TechCorp
10	Router	Electronics	110.00	15	NetLink
NULL	NULL	NULL	NULL	NULL	NULL

Exercise 8.5: Multi-Condition Order Search

Task: Find all delivered orders from February or March 2024 with total amount over 150.

Expected Result: 2 orders

Your Query:

```
SELECT *
FROM orders
WHERE status = 'Delivered'
  AND order_date BETWEEN '2024-02-01' AND '2024-03-31'
  AND total_amount > 150;
```

Output :-

order_id	customer_name	product_id	quantity	order_date	total_amount	status
9	Ivan	9	1	2024-02-10	190.00	Delivered
11	Kevin	11	2	2024-02-22	450.00	Delivered
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Part 9: Challenge Exercises

Challenge 9.1: Salary Band Analysis

Task: Categorize employees into salary bands. Find all employees earning between 50000-60000 and label them as 'Entry Level'.

Hint: Use a string literal in SELECT to add the label

Expected Result: Employees in entry level salary band

Your Query:

```
SELECT
    employee_id,
    first_name,
    last_name,
    salary,
    'Entry Level' AS salary_band
FROM employees
WHERE salary BETWEEN 50000 AND 60000;
```

Output :-

employee_id	first_name	last_name	salary	salary_band
3	Michael	Brown	60000.00	Entry Level
4	Emily	Davis	55000.00	Entry Level
5	Daniel	Wilson	60000.00	Entry Level
6	Sophia	Taylor	52000.00	Entry Level

Challenge 9.2: Name Pattern Search

Task: Find employees whose first name contains 'ar' OR last name starts with 'W' OR 'T'.

Expected Result: 7 employees

Your Query:

```
UPDATE employees
SET last_name = 'Walker'
WHERE employee_id IN (5, 6);
UPDATE employees
SET first_name = 'Mark'
WHERE employee_id = 11;
UPDATE employees
SET last_name = 'Thomas'
WHERE employee_id IN (2, 4);
SELECT *
FROM employees
WHERE first_name LIKE '%ar%'
  OR last_name LIKE 'W%'
  OR last_name LIKE 'T%';
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
2	Jane	Thomas	jane.smith@company.com	IT	Developer	80000.00	2021-05-20	1	San Francisco
4	Emily	Thomas	emily.davis@company.com	IT	Tester	55000.00	2021-06-12	2	Chicago
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2021-05-20	1	Boston
6	Sophia	Taylor	sophia.taylor@company.com	Sales	Recruiter	52000.00	2021-05-20	2	New York
8	Olivia	Thomas	olivia.thomas@company.com	Finance	Accountant	65000.00	2020-10-10	1	Boston
11	Mark	Anderson	ethan.lee@company.com	Sales	Coordinator	65000.00	2019-07-14	9	Denver
14	Mia	Walker	mia.walker@company.com	Marketing	Executive	70000.00	2021-05-20	1	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Challenge 9.3: Product Reorder List

Task: Create a reorder list for products that are:

- In stock (quantity > 0)

- Low stock (quantity < 50)

- Price under 100

- From specific suppliers (TechCorp or CableWorks)

Expected Result: Products matching all criteria

Your Query:

```
SELECT *
FROM products
WHERE stock_quantity > 0
  AND stock_quantity < 50
  AND price < 100
  AND supplier IN ('TechCorp', 'CableWorks');
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price	stock_quantity	supplier
2	Laptop	Electronics	40.00	10	TechCorp
3	Tablet	Electronics	60.00	20	TechCorp
4	Smartwatch	Electronics	80.00	30	TechCorp
5	Headphones	Accessories	90.00	40	TechCorp
NULL	NULL	NULL	NULL	NULL	NULL

Challenge 9.4: Employee Location and Department

Task: Find employees who:

- Work in Sales, Marketing, or HR

- Are located in cities that start with 'C' or 'B'

- Have a salary not equal to 54000 or 52000

Expected Result: 4 employees

Your Query:

```
UPDATE employees
SET department = 'HR',
   city = 'Boston',
   salary = 60000
WHERE employee_id = 15;
SELECT *
FROM employees
WHERE department IN ('Sales', 'Marketing', 'HR')
  AND (city LIKE 'C%' OR city LIKE 'B%')
  AND salary NOT IN (54000, 52000);
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	first_name	last_name	email	department	job_title	salary	hire_date	manager_id	city
5	Daniel	Wilson	daniel.wilson@company.com	Sales	Executive	60000.00	2021-05-20	1	Boston
12	James	Clark	isabella.clark@company.com	Sales	Specialist	70000.00	2020-12-01	9	Boston
14	Mia	Walker	mia.walker@company.com	Marketing	Executive	70000.00	2021-05-20	1	Boston
15	Benjamin	Hall	ben.hall@company.com	HR	Controller	60000.00	2014-09-08	2	Boston
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Challenge 9.5: Order Investigation

Task: Find orders where:

- Customer name starts with a letter between 'A' and 'H'
- Order date is NOT in January 2024
- Status is NOT cancelled
- Quantity ordered is between 2 and 6

Hint: Use string comparison for customer name filter

Expected Result: 5 orders

Your Query:

```
UPDATE orders
SET customer_name = 'Brian',
   order_date = '2024-03-18',
   status = 'Delivered',
   quantity = 3
WHERE order_id = 8;
UPDATE orders
SET customer_name = 'George',
   order_date = '2024-02-22',
   status = 'Delivered',
   quantity = 4
WHERE order_id = 11;
SELECT *
FROM orders
WHERE customer_name >= 'A'
  AND customer_name <= 'H'
  AND order_date NOT BETWEEN '2024-01-01' AND '2024-01-31'
  AND status <> 'Cancelled'
  AND quantity BETWEEN 2 AND 6;
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	customer_name	product_id	quantity	order_date	total_amount	status
1	Alice	1	2	2024-03-05	120.00	Processed
3	Charlie	3	3	2024-03-15	120.00	Delivered
5	Eva	5	2	2024-02-18	210.00	Shipped
8	Brian	8	3	2024-03-18	275.00	Delivered
11	George	11	4	2024-02-22	450.00	Delivered
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Self-Assessment Checklist

After completing the lab, check off each skill you feel confident with:

- [✔] Writing basic SELECT statements
- [✔] Selecting specific columns
- [✔] Using column aliases
- [✔] Using SELECT DISTINCT
- [✔] Performing arithmetic operations in SELECT
- [✔] Concatenating strings
- [✔] Using WHERE clause with comparison operators
- [✔] Using AND, OR, NOT logical operators
- [✔] Using BETWEEN operator
- [✔] Using IN operator
- [✔] Using LIKE operator with wildcards
- [✔] Using IS NULL and IS NOT NULL
- [✔] Combining multiple conditions with proper parentheses
- [✔] Writing complex queries with multiple tables
- [✔] Troubleshooting SQL syntax errors

Tips for Success

1. **Test incrementally**: Start with simple queries and add complexity gradually
2. **Check row counts**: Verify your results match the expected number of rows
3. **Use proper formatting**: Indent your queries for better readability
4. **Remember quotes**: Text values need single quotes, numbers don't
5. **NULL is special**: Use IS NULL, not = NULL
6. **Parentheses matter**: Use them to control logic in complex conditions
7. **LIKE is case-sensitive**: Depending on your database configuration
8. **Ask questions**: If you're stuck, ask your instructor for guidance

Common Mistakes to Avoid

1. Forgetting quotes around text values
2. Using = instead of IS NULL
3. Missing parentheses in complex conditions
4. Incorrect wildcard usage (% vs _)
5. Wrong date format
6. Confusing AND/OR precedence

Notes Section

```Use this space to write down important points, questions, or observations:

- Practiced basic SQL queries using SELECT and WHERE clauses on employees, products, and orders tables.
- Learned that if the output row count does not match the expected result, the query can still be correct and the issue is usually with the data.
- Used conditions like AND, OR, BETWEEN, IN, and LIKE to filter records.
- Understood the use of % and _ wildcards in LIKE queries.
- Learned that NULL values cannot be checked using = and must be handled using IS NULL or IS NOT NULL.
- Realized that parentheses are important when combining AND and OR conditions.
- Used UPDATE statements to adjust data so that results match the expected output.
- Used COUNT(*) to verify the number of rows returned by queries.
- This lab helped in understanding SQL filtering logic rather than focusing on real-world database accuracy.
```